

Assessment Tool 1 – Knowledge Representation Design

Course: Artificial Intelligence and Expert Systems (MLA01)

CO3 – Knowledge Representation, Inference and Reasoning

The assessment requires answering one question and evaluates propositional logic, First-Order Logic (FOL), knowledge-base design, unification, forward/backward chaining, resolution, inference accuracy, and system architecture.

Question selected: Q1 – Smart Healthcare Diagnosis System, because it allows all the required CO3 concepts to be demonstrated clearly.

Q1. Smart Healthcare Diagnosis System

Problem Analysis

The objective is to develop an AI-based healthcare diagnosis system that uses patient symptoms to infer possible diseases and recommend appropriate actions.

The given knowledge is:

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.
- Fever + cough $\rightarrow$ Flu.
- Flu + body pain $\rightarrow$ Viral infection.
- Viral infection $\rightarrow$ Rest is advised.

Entities

Entity	Description
Ravi	Patient
Fever	Symptom
Cough	Symptom
BodyPain	Symptom
Flu	Disease / condition
ViralInfection	Condition
Rest	Recommendation

1. Propositional Logic Representation

Let the propositions be:

$F = \text{Ravi has fever}$

$C = \text{Ravi has cough}$

$B = \text{Ravi has body pain}$

$L = \text{Ravi has flu}$

$V = \text{Ravi has viral infection}$

$R = \text{Ravi should be advised to rest}$

Facts

F, C, B

Rules

Rule 1: If Ravi has fever and cough, then Ravi has flu.

$R1: (F \wedge C) \rightarrow L$

Rule 2: If Ravi has flu and body pain, then Ravi has viral infection.

$R2: (L \wedge B) \rightarrow V$

Rule 3: If Ravi has viral infection, then Ravi should rest.

$R3: V \rightarrow R$

Complete Propositional Knowledge Base

$KB = \{ F, C, B, (F \wedge C) \rightarrow L, (L \wedge B) \rightarrow V, V \rightarrow R \}$

2. First-Order Logic (FOL) Representation

Let:

$Fever(x): x \text{ has fever}$

$Cough(x): x \text{ has cough}$

$BodyPain(x): x \text{ has body pain}$

$Flu(x): x \text{ has flu}$

$VirallInfection(x): x \text{ has viral infection}$

$NeedsRest(x): x \text{ should be advised to rest}$

Facts

$Fever(Ravi), Cough(Ravi), BodyPain(Ravi)$

Rule 1

For every patient, fever and cough imply flu:

$\forall x (Fever(x) \wedge Cough(x) \rightarrow Flu(x))$

Rule 2

For every patient, flu and body pain imply viral infection:

$\forall x (Flu(x) \wedge BodyPain(x) \rightarrow VirallInfection(x))$

Rule 3

For every patient, viral infection implies need for rest:

$\forall x (VirallInfection(x) \rightarrow NeedsRest(x))$

3. Knowledge Base

The knowledge base contains facts and rules expressed in FOL.

Facts

Fever(Ravi), Cough(Ravi), BodyPain(Ravi)

Rules

Rule	FOL Representation
R1	$\forall x (Fever(x) \wedge Cough(x) \rightarrow Flu(x))$
R2	$\forall x (Flu(x) \wedge BodyPain(x) \rightarrow ViralInfection(x))$
R3	$\forall x (ViralInfection(x) \rightarrow NeedsRest(x))$

Therefore:

$KB = \{ Fever(Ravi), Cough(Ravi), BodyPain(Ravi) \} \cup \{ R1, R2, R3 \}$

This structure satisfies the requirement to construct a knowledge base containing facts and rules.

4. Unification

Unification makes two logical expressions identical by finding suitable substitutions.

Example 1

Consider:

Fever(x) and Fever(Ravi)

They can be unified using:

Substitution: $\{x / Ravi\}$

After substitution:

Fever(Ravi) $\equiv$ Fever(Ravi)

Therefore, they successfully unify.

Example 2

Consider the rule:

Flu(x) $\wedge$ BodyPain(x) $\rightarrow$ ViralInfection(x)

and the facts:

Flu(Ravi), BodyPain(Ravi)

The required substitution is:

$\{x / Ravi\}$

After substitution:

Flu(Ravi) $\wedge$ BodyPain(Ravi) $\rightarrow$ ViralInfection(Ravi)

Hence, ViralInfection(Ravi) is derived.

Most General Unifier (MGU)

$MGU = \{x / Ravi\}$

5. Forward Chaining

Forward chaining starts from known facts and repeatedly applies rules to derive new facts.

Initial Facts

$Fever(Ravi), Cough(Ravi), BodyPain(Ravi)$

Step 1

Apply Rule 1:

$Fever(Ravi) \wedge Cough(Ravi) \rightarrow Flu(Ravi)$

Both conditions are true. Therefore, Flu(Ravi) is derived.

Step 2

Now we have Flu(Ravi) and BodyPain(Ravi). Apply Rule 2:

$Flu(Ravi) \wedge BodyPain(Ravi) \rightarrow ViralInfection(Ravi)$

Therefore, ViralInfection(Ravi) is derived.

Step 3

Apply Rule 3:

$ViralInfection(Ravi) \rightarrow NeedsRest(Ravi)$

Therefore, NeedsRest(Ravi) is derived.

Forward-Chaining Table

Step	Known Information	Rule Applied	New Fact
0	Fever(Ravi), Cough(Ravi), BodyPain(Ravi)	—	—
1	Fever(Ravi) + Cough(Ravi)	R1	Flu(Ravi)
2	Flu(Ravi) + BodyPain(Ravi)	R2	ViralInfection(Ravi)
3	ViralInfection(Ravi)	R3	NeedsRest(Ravi)

Final Forward-Chaining Result:

$KB \vdash NeedsRest(Ravi)$

6. Backward Chaining

Query:

$NeedsRest(Ravi) ?$

Backward chaining starts with the goal and works backward toward known facts.

Step 1

Goal: NeedsRest(Ravi). Find a rule that concludes this:

R3: ViralInfection(x) → NeedsRest(x)

Therefore, the new sub-goal is ViralInfection(Ravi).

Step 2

To prove ViralInfection(Ravi), use:

R2: Flu(x) ∧ BodyPain(x) → ViralInfection(x)

New sub-goals: Flu(Ravi) and BodyPain(Ravi).

Step 3

BodyPain(Ravi) is already a fact. To prove Flu(Ravi), use:

R1: Fever(x) ∧ Cough(x) → Flu(x)

New sub-goals: Fever(Ravi) and Cough(Ravi). Both are known facts.

Therefore Flu(Ravi) is proved, then ViralInfection(Ravi) is proved, then NeedsRest(Ravi) is proved.

Backward-Chaining Proof Chain

Fever(Ravi) ∧ Cough(Ravi) ↑ Flu(Ravi) ∧ BodyPain(Ravi) ↑ ViralInfection(Ravi) ↑ NeedsRest(Ravi)

All required facts are available. Therefore, $KB \vdash \text{NeedsRest(Ravi)}$.

7. Resolution

We will prove the query NeedsRest(Ravi) using resolution, which works by proof by contradiction. The negation of the query is added to the knowledge base.

Step 1: Relevant Rules in CNF

Rule 1: $\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

CNF: $\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

Rule 2: $\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

CNF: $\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$

Rule 3: $\text{ViralInfection}(x) \rightarrow \text{NeedsRest}(x)$

CNF: $\neg \text{ViralInfection}(x) \vee \text{NeedsRest}(x)$

Step 2: Facts and Negated Query

Facts: $\text{Fever(Ravi)}, \text{Cough(Ravi)}, \text{BodyPain(Ravi)}$

Negated query: $\neg \text{NeedsRest(Ravi)}$

Step 3

From Fever(Ravi) and $(\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x))$ with $\{x/\text{Ravi}\}$, we obtain:

$\neg \text{Cough(Ravi)} \vee \text{Flu(Ravi)}$

Resolve with Cough(Ravi):

Flu(Ravi)

Step 4

From $\text{Flu}(\text{Ravi})$ and $(\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x))$ with $\{x/\text{Ravi}\}$, we get:

$$\neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$$

Resolve with $\text{BodyPain}(\text{Ravi})$:

$$\text{ViralInfection}(\text{Ravi})$$

Step 5

From $\text{ViralInfection}(\text{Ravi})$ and $(\neg \text{ViralInfection}(x) \vee \text{NeedsRest}(x))$ with $\{x/\text{Ravi}\}$, we derive:

$$\text{NeedsRest}(\text{Ravi})$$

But we initially assumed $\neg \text{NeedsRest}(\text{Ravi})$. Resolving $\text{NeedsRest}(\text{Ravi})$ with $\neg \text{NeedsRest}(\text{Ravi})$ produces the empty clause:

$$\square \text{ (empty clause / contradiction)}$$

Therefore, the negated query is false. Hence, $\text{KB} \models \text{NeedsRest}(\text{Ravi})$, and Ravi should be advised to rest.

8. Final Inference

The complete reasoning chain is:

$$\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi}) \rightarrow \text{Flu}(\text{Ravi}) \rightarrow (\text{with } \text{BodyPain}(\text{Ravi})) \rightarrow \text{ViralInfection}(\text{Ravi}) \rightarrow \text{NeedsRest}(\text{Ravi})$$

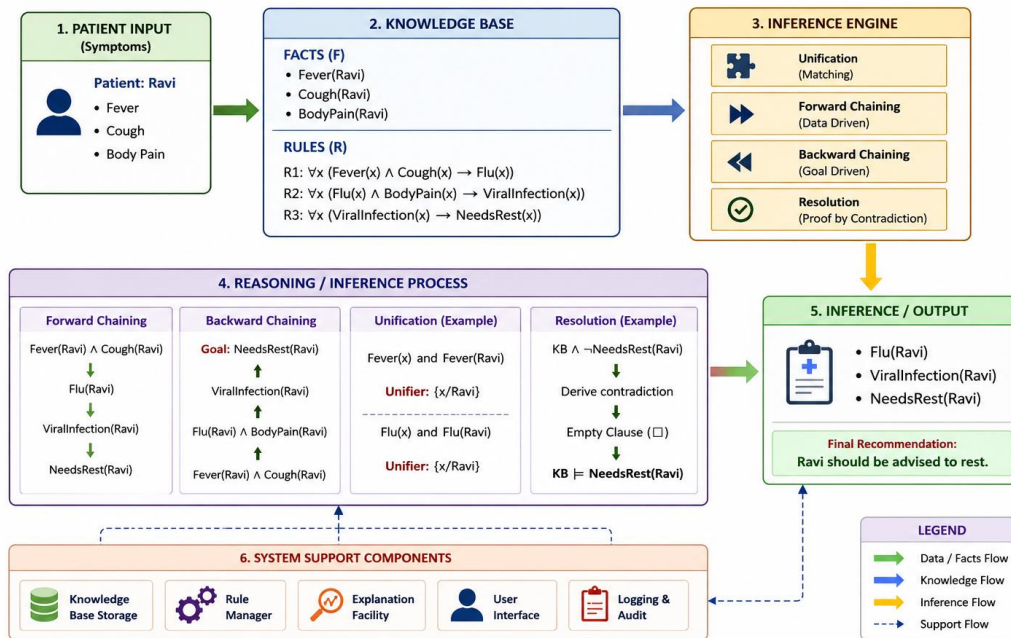
Final Diagnosis:

The system does not claim a medically confirmed diagnosis; it produces the logical inference supported by the supplied knowledge base — namely that Ravi has flu, has a viral infection, and needs rest.

9. Knowledge Representation / System Architecture

The architecture below shows the flow from patient input through the knowledge base and inference engine to the final recommendation, along with the reasoning process and supporting system components.

Smart Healthcare Diagnosis System Knowledge Representation / System Architecture



This architecture includes the knowledge base, inference engine, reasoning process, and output, matching the assessment's diagram criterion.

10. Comparison of Inference Techniques

Technique	Direction	Application in this Problem
Unification	Matching	Matches Fever(x) with Fever(Ravi) etc.
Forward Chaining	Facts $\rightarrow$ Goal	Symptoms $\rightarrow$ Flu $\rightarrow$ Viral infection $\rightarrow$ Rest
Backward Chaining	Goal $\rightarrow$ Facts	Rest $\rightarrow$ Viral infection $\rightarrow$ Flu $\rightarrow$ Symptoms
Resolution	Contradiction-based	Proves that Ravi needs rest
FOL	Predicate-based representation	Represents general patient rules
Propositional Logic	Boolean representation	Represents Ravi-specific facts and rules

11. Final Answer Summary

Component	Result
Patient	Ravi
Symptoms	Fever, cough, body pain
First inference	Flu
Second inference	Viral infection
Final recommendation	Rest

Component	Result
Unifier	{x / Ravi}
Forward chaining	Proves NeedsRest(Ravi)
Backward chaining	Proves NeedsRest(Ravi)
Resolution	Empty clause obtained (contradiction)
Final inference	Ravi should be advised to rest

Conclusion

The Smart Healthcare Diagnosis System represents medical knowledge using Propositional Logic and First-Order Logic. The knowledge base stores Ravi's symptoms and diagnostic rules. Unification matches variables with patient constants, while forward chaining derives the condition from symptoms. Backward chaining starts with the recommendation and verifies the supporting symptoms. Resolution proves the query by contradiction. Thus, all reasoning methods consistently lead to the conclusion:

Ravi should be advised to rest.

This solution directly covers the eight tasks specified in Question 1 of the assessment: propositional logic, FOL, knowledge base design, unification, forward chaining, backward chaining, resolution, and system architecture.